

HOMework 1

46-921

Due Thursday, September 10 at 9:30 AM EDT

Part 1

Revisit the lecture notes, page 15 of Part 1. Row 278303 of the data set appears as extreme in multiple variables. Can you find an explanation for this?

Part 2

In this assignment, you will download some data from the *Wharton Research Data Services (WRDS)* web site and format it. There is a document on the course Canvas page with login details. Read the *WRDS* terms and conditions before using their web site.

While you are logged into the *WRDS* web site, you can look around to see what data are available.

Be sure to submit a Jupyter notebook file with all of your analysis. Also submit the original data file that you downloaded.

After you log in to the *WRDS* site, under the **Get Data** tab choose the **CRSP** database, then **Stock/Security Files**, and then **Monthly Stock File**. Here you will be able to choose what you want to download. For the dates (Step 1), choose September 2000 through June 2015.

For Step 2, choose **TICKER**. This is the ticker symbol. Under “Select an option for entering company codes,” type

BBY MDSY

in the **Company Code** box. (This gets the data for Best Buy and ModSys International.)

In Step 3, choose the following items in “Identifying Information”: **Cusip**, **Company Name**, and **Ticker**. In “Time Series Information,” choose **Price**, **Ask or High**, **Bid or Low**, **Closing Bid**, and **Closing Ask**. In “Share Information,” choose **Number of Shares Outstanding**. Skip “Delisting Information.” In “Distribution Information,” choose **Cumulative Factor to Adjust Price**. Skip “NASDAQ Information” and “Market Information.”

In Step 4, you choose three formats for the output. Select the ***.csv** format. Don’t bother to compress the data. Dates in the form **YYMMDDn8**. are easy to sort. If you wish, you may enter an email address so that they can notify you when the request has been filled. After you “**SUBMIT QUERY**”, you will probably need to wait several seconds for the response to appear. You may be shown some information indicating that the query is being processed. If you are told that no matches were found, you must have skipped one of the above steps (or added a step that isn’t

listed).

Once the request is filled, you will be told the name of a data file that you can download.

Use Python to organize the data set as described below. When you submit your assignment, be sure to answer the **boldface** items.

1. The data file will contain a header telling you which columns correspond to which variables. **How many records, i.e. (stock, day) combinations, does the downloaded data file contain, and how many variables (columns) are there?** Make sure that you understand what all of the variables are, even if you have to go back to *WRDS* and look at the variable descriptions.
2. **We requested two different TICKERs. How many distinct TICKER values appeared in the data file?** The ticker symbol for a company's stock can change over time, and the same ticker symbol can be reassigned to a new company after it has been abandoned by another company. A universal unique identifier of stocks is "CUSIP," named for the Committee on Uniform Security Identification Procedures. **How many distinct CUSIP values appeared in the data file?**
3. There are several possible reasons for extra tickers and/or CUSIPs to appear in a data request. Here, we will diagnose the different TICKERs associated with the CUSIP M7037810. **Figure out why the TICKER changed when it did.** You will need to look elsewhere (outside of *WRDS*) to verify your guess. The COMNAM, Company Name, might help in this task.
4. The "cumulative factor to adjust price" (CFACPR) allows us to make price comparisons over long periods of time. **How many different values of the factor exist in the data file?** If we divide recorded prices PRC by CFACPR, the prices are adjusted for stock splits and other events. **Before dividing, check to see if there are any records for which the factor is missing (NA) but the price exists.** Compute a new data item (PRCadj) for each record by dividing all prices by CFACPR, but preserve the original price PRC as well. **Give an example of a stock and a day on which an adjustment had to be made (the factor is not 1).**
5. Set all undefined or "not available" adjusted prices to appropriate missing values. **How many "not available" adjusted prices are there?**
6. Determine the set of unique dates in the data set. **How many are there?**
7. Negative price values correspond to certain bits of information being missing, but their absolute values still carry useful price information. **Look up the trading volume data on WRDS for the negative prices in the year 2014 in our data set. What common feature do you notice for the negative prices?** Do a Google search to try to determine how the negative price values can be interpreted. For the remainder of the assignment, convert all negative prices to their absolute values. Do this *both* for PRC and PRCadj.

8. Were there any cases in which the Closing Bid (BID) is larger than the Closing Ask (ASK)? If so, offer an explanation for why that happened.

9. We are interested in January 2001 through June 2015. The extra dates in 2000 were chosen so that we could compute one of the popular momentum measures, namely *3-month return*. For each stock, and each month starting in January 2001, compute the 3-month return (call it **momentum**) as

$$\frac{\text{Adjusted price from previous month}}{\text{Adjusted price from four months earlier}} - 1.$$

For example, for January 2001, take the ratio of the 31 December 2000 `PRCadj` to the 30 September 2000 `PRCadj`, then subtract 1. This gives the return over the three months that ended on 31 December 2000. We associate this value with January of 2001, because that is when the number becomes available to use.

10. Construct an appropriate plot to compare the distribution of **momentum** between two sets of months: Those where the price increases in the following month versus those where the price decreases in the following month. Construct a separate plot for the two stocks. **Comment on any structure that you see.**